RadeonShift AI

Accelerating the MI300X Hardware Migration via Generative AI

The Problem

Enterprise AI workloads are facing massive bottlenecks due to a reliance on legacy NVIDIA CUDA codebases. Manually migrating a 50,000-line CUDA codebase to ROCm takes 6+ months of specialized engineering time.

The Solution

RadeonShift AI is a deterministic MoA (Mixture of Agents) pipeline that translates CUDA to ROCm instantly. By leveraging `hipify-perl` for deterministic translation and Fireworks AI's `deepseek-v4-flash` for high-level architectural optimization, it removes manual engineering bottlenecks.

Market & Opportunity

• **TAM**: $25B+ (Global AI Hardware & Software Services Market)
• **SAM**: $8B (Enterprise AI Teams)
• **SOM**: $500M (Active CUDA migration targets in the next 18-24 months)

Competitive Advantage

• **5,000x ROI** over human engineering hours
• Parallel MoA execution (NVIDIA Purist + AMD Optimizer)
• Anchored entirely on ROCm Tooling (`hipify-perl`, `rocm-smi`) rather than prone-to-hallucinate zero-shot LLM prompts.

Team & Tech

Built natively with React 19 / Next.js 16 and a Python/FastAPI backend utilizing live AMD Instinct MI300X cloud tooling.